Data Management and Sharing Plan

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NSF AGS-PRF Application

Products of Research

The project will produce source code, trained-model configuration files, evaluation scripts, derived hindcast diagnostics, summary tables, selected nonrestricted figures, metadata, teaching materials, and manuscripts. The project will use third-party climate datasets such as PRISM and ERA5. Those source datasets will not be redistributed if license or provider terms require users to obtain them from the original sources.

Data and Code Formats

Code will be released in plain-text Python and shell scripts with environment files where possible. Derived data products will be released in standard scientific formats such as NetCDF, NPZ, CSV, or GeoTIFF as appropriate, with README files describing variables, units, grids, time coverage, cross-validation splits, and known limitations. Figures and tables used in manuscripts will be preserved with scripts or notebooks sufficient to reproduce them from derived products.

Access and Sharing

Nonrestricted derived products and code will be shared through a public repository and archived with a persistent DOI through a repository such as Zenodo or an institutional archive. Large intermediate files and third-party raw data will be documented with scripts and instructions so authorized users can reproduce them from original providers. Repository documentation will distinguish between raw third-party inputs, derived hindcast products, trained-model outputs, and manuscript-ready diagnostics.

Preservation

Final code, metadata, manuscripts, teaching materials, and derived data needed to reproduce major figures and tables will be preserved for at least five years after the end of the award. Versioned releases will be used for manuscript-associated artifacts so that published results can be traced to a specific code and data snapshot.

Privacy, Security, and Ethical Considerations

The project uses environmental and climate data, not human-subjects data. No personally identifiable information will be collected. Model limitations, uncertainty, and appropriate-use caveats will be documented to reduce the risk of overinterpreting experimental forecast products. Experimental forecasts will be presented as research outputs and not as official operational guidance.

Roles and Responsibilities

The Fellow will maintain the repository, prepare metadata, document workflows, and archive final products. The sponsoring scientist will advise on scientific quality control, reproducible workflow design, and appropriate dissemination. Data-management tasks will be reviewed at manuscript and release milestones to ensure that code, metadata, and derived products remain consistent.

Workflow Documentation and Reuse

The project will include documentation for environment setup, data preprocessing, model training, evaluation, and figure generation. Where third-party raw data cannot be redistributed, the repository will provide clear instructions for obtaining the source data and rebuilding derived products. Configuration files will record variables, pressure levels, spatial grids, lead times, training years, validation years, and model settings.

Quality Control for Shared Products

Before release, derived products will be checked for missing values, unit consistency, grid metadata, time-coordinate accuracy, and agreement with manuscript figures. Public products will include caveats explaining that the forecasts are research hindcasts rather than operational warnings. This documentation will help future users understand the scope, limits, and appropriate interpretation of the project outputs.